

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I Pravin R / 192421292 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

Solution:

Bayes' Theorem (Naïve Bayes)

Given:

$$P(A) = 1/4$$

$$P(B) = 1/2$$

$$P(C) = 1/4$$

$$P(\text{Blue}|A) = 3/10$$

$$P(\text{Blue}|B) = 6/15 = 2/5$$

$$P(\text{Blue}|C) = 2/15$$

Step 1: Calculate the total probability of selecting a blue car.

$$P(\text{Blue}) = P(A)P(\text{Blue}|A) + P(B)P(\text{Blue}|B) + P(C)P(\text{Blue}|C)$$

$$= (1/4 \times 3/10) + (1/2 \times 2/5) + (1/4 \times 2/15)$$

$$= 3/40 + 1/5 + 1/30$$

$$= 37/120$$

Step 2: Apply Bayes' Theorem.

$$P(C|\text{Blue}) = [P(C) \times P(\text{Blue}|C)] / P(\text{Blue})$$

$$= [(1/4) \times (2/15)] / (37/120)$$

$$= (1/30) \times (120/37) = 4/37$$

Final Answer:

$$P(C|Blue) = 4/37 \approx 0.1081$$

2. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires).

(5 Marks)

a) Front identified as Front: 119 (TP)

b) Front identifies as Back: 06

c) Back identified as Back: 120

d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Solution:

Given:

Total Tires = 250

Confusion Matrix:

	Predicted Front	Predicted Back
Actual Front	119 (TP)	6 (FN)
Actual Back	5 (FP)	120 (TN)

$$TP = 119$$

$$TN = 120$$

$$FP = 5$$

$$FN = 6$$

1. Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$= (119 + 120) / (119 + 120 + 5 + 6)$$

$$= 239 / 250$$

$$= 0.956$$

$$= 95.60\%$$

2. Precision

$$\text{Precision} = TP / (TP + FP)$$

$$= 119 / (119 + 5)$$

$$= 119 / 124$$

$$= 0.9597$$

$$= 95.97\%$$

3. Sensitivity (Recall)

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$$

$$= 119 / (119 + 6)$$

$$= 119 / 125$$

$$= 0.952$$

$$= 95.20\%$$

4. Specificity

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

$$= 120 / (120 + 5)$$

$$= 120 / 125$$

$$= 0.96$$

$$= 96.00\%$$

5. F1-Score

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$= 2 \times (0.9597 \times 0.952) / (0.9597 + 0.952)$$

$$= 0.9559$$

$$= 95.59\%$$

Final Answers:

Accuracy = 95.60%

Precision = 95.97%

Sensitivity = 95.20%

Specificity = 96.00%

F1-Score = 95.59%

3. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

- a) Diseased identified as Diseased: 138**
- b) Diseased identified as Healthy: 12**
- c) Healthy identified as Healthy: 135**
- d) Healthy identified as Diseased: 15**

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Solution:

Given:

Total Cases = 300

Confusion Matrix:

	Predicted Diseased	Predicted Healthy
Actual Diseased	138 (TP)	12 (FN)
Actual Healthy	15 (FP)	135 (TN)

TP = 138

TN = 135

FP = 15

FN = 12

1. Accuracy

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

$$= (138 + 135) / (138 + 135 + 15 + 12)$$

$$= 273 / 300$$

$$= 0.91$$

$$= 91.00\%$$

2. Precision

Precision = $TP / (TP + FP)$

$$= 138 / (138 + 15)$$

$$= 0.9020$$

$$= 90.20\%$$

3. Sensitivity (Recall)

$$\begin{aligned}\text{Sensitivity} &= \text{TP} / (\text{TP} + \text{FN}) \\ &= 138 / (138 + 12) \\ &= 138 / 150 \\ &= 0.92 \\ &= 92.00\%\end{aligned}$$

4. Specificity

$$\begin{aligned}\text{Specificity} &= \text{TN} / (\text{TN} + \text{FP}) \\ &= 135 / (135 + 15) \\ &= 135 / 150 \\ &= 0.90 \\ &= 90.00\%\end{aligned}$$

5. F1-Score

$$\begin{aligned}\text{F1-Score} &= 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \\ &= 2 \times (0.9020 \times 0.92) / (0.9020 + 0.92) \\ &= 1.6597 / 1.8220 \\ &= 0.9109 \\ &= 91.09\%\end{aligned}$$

Final Answers:

Accuracy = 91.00%
Precision = 90.20%
Sensitivity = 92.00%
Specificity = 90.00%
F1-Score = 91.09%

4. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Solution:

Given:

Total Emails = 400

Confusion Matrix:

	Predicted Spam	Predicted Not Spam
Actual Spam	180 (TP)	20 (FN)
Actual Not Spam	30 (FP)	170 (TN)

TP = 180

TN = 170

FP = 30

FN = 20

1. Accuracy

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

$$= (180 + 170) / (180 + 170 + 30 + 20)$$

$$= 350 / 400$$

$$= 0.875$$

$$= 87.50\%$$

2. Precision

Precision = $TP / (TP + FP)$

$$= 180 / (180 + 30)$$

$$= 0.8571$$

$$= 85.71\%$$

3. Sensitivity (Recall)

$$\begin{aligned}\text{Sensitivity} &= \text{TP} / (\text{TP} + \text{FN}) \\ &= 180 / (180 + 20) \\ &= 180 / 200 \\ &= 0.90 \\ &= 90.00\%\end{aligned}$$

4. Specificity

$$\begin{aligned}\text{Specificity} &= \text{TN} / (\text{TN} + \text{FP}) \\ &= 170 / (170 + 30) \\ &= 170 / 200 \\ &= 0.85 \\ &= 85.00\%\end{aligned}$$

5. F1-Score

$$\begin{aligned}\text{F1-Score} &= 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \\ &= 2 \times (0.8571 \times 0.90) / (0.8571 + 0.90) \\ &= 1.5429 / 1.7571 \\ &= 0.8780 \\ &= 87.80\%\end{aligned}$$

Final Answers:

Accuracy = 87.50%
Precision = 85.71%
Sensitivity = 90.00%
Specificity = 85.00%
F1-Score = 87.80%

